Assessment Details and Submission Guidelines	
Unit Code	**NIT6160**
Unit Title	**Data Warehousing and Mining**
Term, Year	H8W4, 2026
Assessment Type	Project, Individual
Assessment Title	Health Dataset Analysis and Prediction
Purpose of the assessment (with ULO Mapping)	This assignment assesses the following Unit Learning Outcomes; students should be able to demonstrate their achievements in them. 1. Critically evaluate data quality to advocate application of data pre-processing techniques; 2. Devise implementation solutions that involve supervised learning including regression and classification on real-world data sets; 3. Experiment and evaluate unsupervised learning solutions including clustering and association rules mining on large datasets; and 4. Critically reflect on advantages and disadvantages of particular data mining solutions to solve real life problems and evaluate their usefulness and useability.
Weight	50% of the total assessment
Total Marks	100;
Word limit	Minimum 1000 words for the report
Due Date	Demonstration/Viva, Week 7, Session 12, During the lab time. File Submission, Week 8, Monday, 11:59PM
Submission Guidelines	• All work must be submitted on VUC by the due date. • Students are required to submit three files: a document file, the presentation PowerPoint file, and a Python Jupyter Notebook. • The assignment file must be in MS Word format, 1.5 spacing, 11-pt Calibri (Body) font, and 2 cm margins on all four sides of your page with appropriate section headings. The presentation file should be in PowerPoint format. • Reference sources must be cited in the text of the report and listed appropriately at the end in a reference list using the IEEE referencing style. • Late submission penalty: a penalty will be applied for each day of delayed submission.

ASSIGNMENT DESCRIPTION

The objective of this assignment is to conduct a comprehensive machine learning analysis of a health-related dataset to explore various factors associated with health conditions, risk factors, patient outcomes, or disease prevalence, and to develop predictive models for healthcare-related decision making. Students will apply machine learning techniques to analyze the dataset, develop predictive models, and present their findings in a clear and concise manner, including model performance metrics and feature importance analyses.

Students will apply machine learning techniques to:

- Investigate demographic, clinical, behavioural, and lifestyle variables associated with the selected health condition or healthcare outcome
- Build and evaluate predictive models for healthcare-related prediction or classification tasks
- Present results clearly, including model performance metrics (e.g., accuracy, precision, recall, F1-score, ROC-AUC) and feature importance analyses

Task 1 Dataset Selection **[10 marks]**

You will select a **health-related dataset** for analysis. The chosen dataset should ideally (but not necessarily) include variables such as:

- Demographics: age, gender, ethnicity
- Clinical measurements: blood pressure, cholesterol levels, BMI, glucose levels, heart rate, or other relevant medical indicators
- Diagnostic or medical test results: laboratory findings, ECG results, imaging results, or disease-specific test outcomes
- Lifestyle factors: smoking status, physical activity level, diet, alcohol consumption, sleep patterns, etc.
- Health outcomes: disease presence/severity, diagnosis status, recovery outcomes, hospital readmission, or other health-related target variables

Example health-related datasets include heart disease, diabetes, breast cancer, stroke prediction, chronic kidney disease, COVID-19, mental health, obesity, sleep health, and hospital readmission datasets.

Potential sources:

- UCI Machine Learning Repository
- Kaggle
- Public health agencies and government open-data portals
- WHO, CDC, or other healthcare-related repositories

Before proceeding to analysis, show your chosen dataset to the lecturer and obtain confirmation to ensure data quality, completeness, and suitability for your machine learning project.

Task 2 Research Questions [10 marks]

Research questions are essential for guiding the machine learning analysis and exploration of the selected health-related dataset. By considering the dataset's variables and healthcare context, students can uncover insights into disease prevalence, risk factors, patient outcomes, healthcare trends, and treatment effectiveness. These questions will provide a framework for investigating relationships and associations within the dataset, enhancing understanding and supporting evidence-based healthcare decisions and policies.

To build a strong foundation, you should:

1. Conduct a Literature Survey:

Review existing research and literature related to the selected health condition, healthcare problem, or predictive modeling techniques to better understand current knowledge, trends, and research gaps. The literature survey must include a summary of at least three recent research papers from reputable journals or conferences.

2. Formulate Research Questions:

Develop specific research questions based on the dataset's variables and healthcare context. Example questions might include:

- What are the significant factors associated with the selected health condition or outcome?
- Is there a relationship between demographic factors (e.g., age or gender) and the target health outcome?
- How do lifestyle factors, such as diet, exercise, smoking, or sleep, influence health outcomes?
- Which variables contribute most significantly to prediction accuracy in the machine learning model?
- Can machine learning models effectively predict disease risk, hospital readmission, or patient outcomes?
- This task will combine insights from the literature with dataset analysis, supporting a well-rounded approach to answering critical healthcare-related research questions.

Task 3 Data Exploration [10 marks]

The objective of this data exploration task is to understand and summarize the characteristics of the dataset collected in Task 1. Once the relevant data has been collected, the following tasks need to be performed:

- Load and Examine the Dataset: Load the selected dataset and examine its structure, including variable types and any missing data patterns.

- Explore Variable Distributions: Use summary statistics and visualizations to explore the distribution of each variable, identifying any irregularities, trends, or patterns.

- Perform Descriptive Analysis: Conduct descriptive analyses to provide an overview of the data's characteristics, including variable ranges, counts, and key relationships.

- Calculate Central Tendency and Dispersion: Compute key measures of central tendency (e.g., mean, median) and dispersion (e.g., variance, standard deviation) to gain insights into the dataset's features, detect potential outliers, and inform the data pre-processing and modeling steps.

This exploration will provide a foundational understanding of the dataset's structure and variability, supporting informed decisions in the modeling process.

Task 4 Data Visualization **[10 marks]**

Graphical representations play a crucial role in exploratory data analysis, allowing students to visually inspect the dataset, identify patterns and relationships, and generate hypotheses for further investigation. You need to create graphical representations (histograms, box plots, line graphs) to visually depict the distribution of variables and identify any patterns, trends, or outliers. In addition, explore the relationships between variables through scatter plots or correlation matrices to understand potential associations.

Task 5 Machine Learning Models and Performance Evaluation **[20 Marks]**

In this task, students will apply machine learning models to the dataset to develop predictive insights. This process includes selecting, training, and evaluating various models to assess their predictive performance and identify the best model for the disease prediction.

The steps involved in this task are as follows:

a) Model Selection:

Select any 3 appropriate machine learning models to test on the dataset. Suggested models may include:

- Logistic Regression: A simple, interpretable model suitable for binary classification problems.
- Decision Trees and Random Forests: Models that capture complex patterns, useful for handling non-linear relationships.
- Support Vector Machines (SVM): Effective for high-dimensional spaces, useful if the dataset has multiple features.
- Neural Networks: A more complex model option, particularly useful if the dataset is large and has non-linear patterns.

Justify the choice of models based on the dataset characteristics and desired interpretability versus accuracy trade-offs.

b) Data Splitting:

Divide the dataset into training and test sets (e.g., 80-20 split). Consider using cross-validation to ensure model robustness and avoid overfitting. This involves training the model on multiple subsets of the data and averaging the results to obtain more reliable performance estimates.

c) Model Training and Hyperparameter Tuning:

Train each selected model on the training set. For better performance, optimize each model's hyperparameters using techniques like grid search or randomized search cross-validation.

d) Performance Evaluation Metrics:

Evaluate each model's performance on the test set using the following metrics:

- Accuracy: The percentage of correctly classified instances, which provides a basic performance indicator.
- Precision and Recall: Metrics that provide insights into the model's ability to accurately predict positive cases (Precision) and identify actual positive cases (Recall).
- F1 Score: A balanced measure that considers both precision and recall, useful for evaluating models on imbalanced data.

Use confusion matrices to visualize each model's performance and to understand the types of errors it makes.

e) Model Comparison and Selection:

Compare the performance of all models based on the evaluation metrics to determine the most suitable model for predicting health condition Consider both accuracy and interpretability; a simpler model might be preferred if it achieves comparable results to a complex model.

Task 6 Interpretation, Conclusion and References **[15 marks]**

In this final task, students will summarize key findings and interpret the results in relation to the research questions.

- **Summarize Findings:** Provide a clear summary of the main findings from the machine learning analysis, linking them back to the original research questions. Highlight important insights regarding risk factors, predictors, and any associations identified in the dataset.
- **Discuss Implications:** Discuss the implications of these findings for understanding the disease risk factors and potential prevention strategies, noting how these insights may inform health interventions or policies.
- **Identify Limitations:** Acknowledge any limitations in the analysis, such as data quality, sample size, or model constraints. This section should provide a critical evaluation of potential biases or data limitations that may have impacted the findings.
- **Future Research Directions:** Suggest directions for future research that could build on this analysis, such as testing additional variables, exploring other predictive models, or applying the analysis to different populations.
- **References and Citations:** Include relevant references and citations from the literature, ensuring that all sources used throughout the project are properly credited.

Task 7 Presentation/Viva **[25 marks]**

Student will present a concise overview of their project, covering:

- **Introduction and Objectives:** Briefly introduce the project, research questions, and dataset.
- **Dataset and Preprocessing:** Summarize key dataset variables and preprocessing steps.
- **Exploratory Data Analysis (EDA):** Share visualizations and major findings that informed model choices.
- **Modeling and Evaluation:** Describe models used, performance metrics, and reasoning for the final model choice.
- **Interpretation and Implications:** Discuss main insights, limitations, and recommendations for future research.
- **Conclusion:** Recap key takeaways.
- **Q&A Session:** Be ready to answer questions.
  Present clearly, with visual aids. Keep within the time limit (10 minutes).

During the laboratory and tutorial session in Session 12, your lecturer will ask questions related to your project. You are expected to demonstrate your dataset processing and analytical skills. It's essential that students attend and understand the entire assignment and answer the viva questions . Be prepared to demonstrate code execution, explain data preprocessing steps, model choices, and discuss results from the analysis.

FILE SUBMISSION

You need to submit three files in the Assignment Submission Link:

1. Written Report - Submit a written report detailing the analysis, findings, and conclusions. Include visualizations, tables, and statistical outputs to support the analysis.
2. Presentation PowerPoint file
3. A Python Jupyter Notebook file